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Machine learning for sperm selection

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Abstract

Infertility rates have increased worldwide over the last few decades, and more couples are seeking fertility care. Over 2.5 million cycles are being performed every year globally, but the success rate of assisted reproductive technologies (ART) has remained at ~33%. Machine learning, an automated method of data analysis based on patterns and inference, is increasingly being deployed within the health sector to improve both diagnostics and therapeutics. This technique is already aiding some fertility clinics with embryo selection, and recently, has been applied in research laboratories to improve sperm analysis and selection. We see tremendous opportunities for machine learning to advance male fertility. The fundamental challenge of sperm selection – selecting the most promising candidate out of 10^8 gametes – presents a challenge that is uniquely well suited to the high-throughput capabilities of machine learning algorithms paired with modern GPUs. Here we review the emerging field of machine learning for sperm selection and highlight

the areas where the most significant gains are possible, as well as the machine learning strategies best suited to achieve them. We highlight areas where existing clinical machine learning approaches can be readily applied to male fertility, as well as aspects – both technical and social – that are emerging.

Key points:

- Selection of sperm during ICSI is currently not standardized as the WHO guidelines are interpreted subjectively by clinical experts.
- Machine learning (ML) shows promise in improving ICSI by guiding the clinicians to objectively select the sperm.
- Most reported ML methods for sperm head classification rely on data labeled by clinical experts, despite the variability that exists between them.
- Recent studies on single sperm motility and DNA integrity can be levered as un-biased quantitative measures for ML algorithm-based sperm selection.
- Considerations of data quality, subjectivity in training data, and representativeness of data are crucial to minimize bias.
- Ethical questions such as data privacy, control over the decision-making process, and data variability warrant further consideration.

Introduction

The rise in infertility is a serious health concern. Globally, around 50 million couples suffer from fertility-related issues of which 30% are solely male factors.^{1,2} Environmental pollution,³⁻⁵ unhealthy diet,^{6,7} and obesity^{8,9} have been associated with reduced sperm concentration and motility, while demographic and geographic factors and subsequent cultural diversities also

contribute to the infertility issue and its global prevalence.^{10–12} Assisted reproductive technologies (ART) have been used increasingly over the past few decades, with over 2.5 million cycles now being performed annually.¹³ However, despite the increasing number of ART cases each year,¹⁴ the success rate of ART has remained at ~33% per cycle.¹⁵

The quality of gametes is crucial to the success rate of the ART cycle. During an ART cycle, the majority of extracted oocytes from the female are fertilized and used. They are either used for embryo transfer in the same cycle or they are frozen for use in subsequent cycles. However, the male side involves selection of only a few sperm (on the order of 10) from many (on the order of 10^8) that are genetically heterogeneous, representing a demanding sorting challenge. In *in-vitro* fertilization (IVF), a selected subpopulation of ~50,000 sperm is placed near an oocyte to initiate fertilization. In contrast, intracytoplasmic sperm injection (ICSI) requires careful manipulation of an individually selected sperm, which is then injected directly into the oocyte by an embryologist.

Clinical procedures for sperm selection typically involve the removal of dead cells and debris via centrifugation (i.e. density gradient centrifugation, DG),^{16,17} separation of highly-motile sperm through sperm washing media (i.e. swim-up assay, SU)¹⁸ or a combination of both.^{19,20} Subsequent to the primary selection, a sample with ~1% of the original sperm concentration is retrieved, from which an aliquot is taken for the ART cycle. That is, for a normal patient with semen concentration near the World Health Organization (WHO) standard (>15 million sperm/mL),²¹ the resulting concentration of sperm is ~150,000/mL. While this number is enough for an IVF procedure – which generally requires ~ 50,000 sperm²² – an ICSI process requires another ~ 10^5 fold narrowing to choose a single sperm to be injected directly into an egg. Therefore, an aliquot on the order of ~10 μ L is taken from selected subpopulation, with the number of sperms retrieved on the order of ~ 10^3 . However, due to time constraints during the ICSI procedure, it is not practical for a clinician

to individually examine 10^3 sperm. In practice, a small fraction of these sperm is observed prior to manual selection based on World Health Organization (WHO) criteria,²³ and that final selection process is clinician-specific and highly subjective. Despite recent technological developments that provide alternatives to conventional clinical methods,^{14,24} sperm selection methods still need to be developed further to include other important ART outcome characteristics such as fertilization rate and live-birth outcome.

Currently, the WHO morphology assessment criteria provide the most reliable source to guide clinician's sperm selection. The subjective nature of the interpretation of these criteria results in a lack of standardization. Besides morphology, both motility and DNA integrity are frequently used to monitor sperm quality prior to the application of ART, but there is a lack of data available at the single sperm level. Motility is currently assessed at the sample level via computer-assisted sperm analysis (CASA) based on the motion of hundreds of sperms, and then assessed visually by the embryologist on the day of treatment. Current DNA integrity tests determine the percentage of sperm with fragmented DNA in a semen sample, but DNA quality is not currently implemented as a metric for sperm selection on the day of an ART cycle, due to the invasive nature of DNA analysis assays. Assessing all three parameters of morphology, motility and DNA integrity – either directly or via a trained algorithm – could greatly inform and standardize the selection of single sperm for ICSI.

Recent developments in machine learning algorithms have enabled a plethora of applications in computer vision,^{25,26} digit recognition,²⁷ health monitoring and diagnostics,^{28–32} drug discovery,^{33,34} language processing,^{35,36} and many others (BOX 1). ML has been used to process a large amount of data in biomedical fields including fertility and reproductive medicine,^{37,38} and has shown the potential to improve current ART practices both in the clinic and through research.

In the clinic, machine learning can help clinicians make more informed and consistent decisions in sperm selection. The capabilities of machine learning algorithms to process large amounts of data provides the potential to establish new links between sperm quality metrics at the single-cell level which may further improve sperm selection capabilities in the future. Furthermore, an algorithm originally trained on sperm data can be later enriched with other outcomes such as fertilization rate, blast formation rate or live birth rate.

In this review, we discuss how machine learning capabilities can advance sperm selection. Our focus is on the near-term achievable machine learning methods based on available assessments of selected sperm (i.e. morphology, motility, and DNA integrity, **Figure 1**) that are akin to those used to train clinicians. We start by highlighting some of the previous works that have employed various machine learning strategies in classifying and assessing sperm morphology. Then we discuss the importance of studying clinically-accessible features that distinguish individual sperm - such as morphology, DNA integrity and motility – to training a smarter algorithm for the best informed and standardized sperm selection in assisted reproduction. Finally, we point out the necessary considerations when implementing machine learning strategies in clinical settings including potential issues of data bias and associated ethical questions.

Target cohort for machine learning sperm selection technologies

The patients that would benefit most from machine learning based sperm selection technologies are those who require the ICSI procedure. ICSI is the most common method of fertilization used for ART. In the United State alone, 908,767 ICSI cycles were performed between 1996 and 2012, corresponding to 65.1% of all ART cycles.³⁹ The use of ICSI increased from 36.4 to 76.2%

between 1996 and 2012. Also, according to ART data compiled by Dyer et al., at least 60% of non-donor ART cycles have employed ICSI globally.⁴⁰

ICSI is especially important for couples that suffer from male factor infertility. According to the study by Boulet et al., about 90% of IVF cycles with male factor infertility use ICSI procedures.³⁹ Male factor infertility can include azoospermia (complete absence of sperm in ejaculate), oligozoospermia (low sperm count in ejaculate), globozoospermia (absence of acrosome), asthenozoospermia (no motile sperm in ejaculate), teratozoospermia (<5% morphologically normal sperm in ejaculate) or patients with normal sperm parameters but high sperm DNA fragmentation. Among these, 10-15% are azoospermic and sperm must be obtained percutaneously or via surgery. In those cases, the challenge is locating a single sperm, or a few sperm, within a complex sample and machine learning based selection is not as helpful. However, for the other 85-90% of cases, individual sperm must be selected from a pool and there is an opportunity for machine learning to aid clinicians in that selection process. Even for severe male infertility samples with relatively low sperm count, a well-trained ML algorithm can inform the clinical sperm selection process. To provide the maximum benefit for patients and clinicians, algorithms must be trained on all established sperm quality metrics, including those accessible during clinical selection (e.g. motility, morphology), and those that are not (e.g. DNA quality). In the case of DNA integrity, algorithms can learn proxies for this metric, and apply those learnings in clinical sperm selection. In this sense, breadth is important in the training of algorithms, in direct analogy with the training of clinicians. Patients and clinicians stand to benefit greatly from richly trained, and carefully applied, machine learning algorithms.

BOX 1 | Machine learning

Machine learning is an automated data analysis method whereby an algorithm predicts the outcome of an unseen event based on patterns and inference identified in input data, without providing explicit instructions.^{41–44} Most machine learning methods are classified into supervised and unsupervised learning. In supervised learning, the algorithm is trained with a set of labelled data (in which each input datum is paired with an outcome) to identify the input-output correlation and then use this correlation to predict the outcome for new cases. Supervised learning algorithms can only make a prediction based on labeled training data that still requires human intervention. Binary classification is a typical example of supervised learning, in which input data (i.e. sperm morphology) are classified into one of the two possible categories (i.e. normal versus abnormal sperm). In unsupervised learning, the algorithm identifies underlying and maybe previously unknown patterns in unlabelled input data and with minimal human intervention. An important example of unsupervised learning is “clustering” whereby the algorithm determines shared characteristics of the input data set to group the data and extrapolate patterns for future classification of similar data sets.

Deep learning is a machine learning technique based on an artificial neural network (ANN) that leverages multiple levels or layers which can interact with each other to arrive at the decision, resembling how a biological brain works. Owing to its capability to handle massive data, deep learning has been exploited in various research fields such as medicine, language recognition, processing of text documents, and others.^{45–47} When the number of layers is large and thus the interaction between each layer becomes complex, the neural network is referred to as a Deep Neural Network (DNN). In contrast to traditional machine learning, deep learning exploits the highly complex interactions between each layer to extract useful information from data. When a

convolution operation is employed, the DNN is said to be a Convolutional Neural Network (CNN), and this form of deep learning has been very effective in cell image classification.

[BOX 1-END]

BOX 2 | Supervised learning algorithms for image analysis

A suite of machine learning techniques can be employed to expedite, automate, and standardize image analysis. A support vector machine (SVM) enables non-linear mapping of manually-extracted features of an image (e.g. area, perimeter, Fourier descriptors, Hu moments) to different categories.⁴⁸ A dictionary learning approach involves training class-specific dictionaries of smaller image patches such that images can be approximated as a linear combination of smaller elements.⁴⁹ Deep learning approaches – such as convolutional neural networks (CNNs), autoencoders, and recurrent neural networks (RNNs) – have proven most successful when applied to image analysis. CNNs are a class of deep neural networks designed to extract locally-correlated features via multiple layers of trained convolutional kernels and artificial neurons that map entire images to different categories. Autoencoders are trained to copy the image input to its output while preserving only the relevant aspects of the image (e.g. for image segmentation).⁵⁰ RNNs process sequential images (i.e. video data) by selectively passing information across sequence steps, learning both short-term and long-term dependencies.

[BOX 2-END]

Machine learning strategies for sperm morphology assessment

Current morphology analysis

Morphology is an important parameter in characterizing sperm quality, and the main selection metric applied in current ART practices. The fundamental challenge of classifying sperm morphology is dictated by biology as human sperm appears in a wide range of sizes and shapes.⁵¹ Sperm morphology is currently assessed by clinicians with reference to their experience and the WHO criteria (**Table 1**), according to which a semen sample is considered normal if it has > 4% morphologically normal sperm.^{23,51}

The WHO guidelines have changed over time, and became stricter with more precise descriptions of normal forms. The criteria of ‘normal’ sperm morphology has been defined based on sperm retrieved from the female reproductive tract as well as the surface of zona pellucida (an extracellular protein matrix surrounding the egg).²³ Initially, evaluation of sperm morphology was simply identifying a sperm as normal if no obvious abnormalities such as a tapered head, absence of an acrosome, or a thick midpiece were observed.⁵¹ However, a ‘strict’ criteria⁵² was later developed due to the poor reported correlation between liberal criteria and the ability to conceive.⁵³ The strict criteria provides a detailed and accurate description of normal size and shape as well as possible abnormalities for each sperm head, neck, midpiece and tail.

However, the subjective interpretation of criteria has made it difficult for embryologists to achieve both inter-expert and inter-laboratory consistency on morphology.^{53–55} Using blind assessments, Brito showed a significant variability on both intra- and inter-laboratory analysis of bovine sperm morphology.⁵⁴ Barroso et al. reported a low level of inter-laboratory agreement on manual assessment of sperm morphology.⁵⁵ In addition, differences in experience levels among clinicians can also be a source of variability.⁵⁶

Sperm head staining is often used to better visualize the head during morphology assessment. Numerous methods including Papanicolaou (PAP),⁵⁷ Rapidiff (RD),⁵⁷ Testsimplets,⁵⁸ Diff-Quik,⁵⁹

SpermBlue (SB)⁶⁰ or Shorr staining⁶¹ have been developed to better assess a semen sample. These methods aid the embryologists in determining a sperm as “normal” or “abnormal” by enhancing the color contrast of the sperm head from the background. However, as these methods rely on staining, several comparative studies have argued that the measured dimension of sperm head varies considerably between different staining methods.^{61–64} Maree et al. compared the effect of three staining methods (PAP, RD and SB), and observed that the morphometric dimensions for human sperm were different between the methods.⁶² On the other hand, Henkel et al. concluded that while results obtained using PAP and Shorr staining agree to a great extent, the Testsimplets staining method yields results that are statistically different from the other two.⁶¹ The sperm heads were stained for morphology assessment after air-drying or fixation, which directly impacts sperm vitality and motility. Stained sperm cannot be used in ICSI, given the need for live and motile sperm.

One challenge in the application of current morphology assessments is that a semen sample is labeled as “normal” if it passes the 4% threshold, despite very large variability at the cell level. While this variability is less of an issue for IVF or intrauterine insemination (IUI) procedures where ultimate selection employs the natural egg-sperm interactions, variability poses a significant challenge for an ICSI procedure. Despite the fact that the number of sperm is significantly reduced after washing and processing steps such as density gradient centrifugation (DGC) and/or swim-up, there are still $10^2 \sim 10^3$ sperms in the processed subpopulation. Given the short timeframe of an ICSI procedure, selecting a single sperm is extremely challenging and subject to inter-expert and intra-clinic variability.

Machine learning approaches for sperm classification based on morphology

The variability in sperm morphology assessment results has motivated the development of computer software for standardized sperm analysis such as FERTECH.⁶⁵ FERTECH is an image-processing software that classifies a 100× magnification image of a sperm cell according to strict criteria. The software identifies all objects in an image, removes the background, extracts the sperm head shape, and evaluates the shape based on strict criteria. Later, the same group reported an automated system called Integrated Visual Optical System (IVOS) incorporating an optical setup and image digitizing and analysis setup. The system enables automated analysis of an entire sperm smeared slide.⁶⁶ When compared to manual analysis, the evaluation performance of the IVOS system is as good as that of the clinical experts⁶⁷ and shows high repeatability.⁶⁸ However, the way these systems score a sperm as normal or abnormal is strictly based on the assessments done by human experts according to WHO criteria, which are known to have bias. Moreover, the performance of these two systems have shown to depend on the applied staining method.^{55,69,70}

The development of machine learning over the last decade has provided opportunities to standardize sperm morphology assessment and add new classification capabilities based on morphological defects. (Figure 2) For instance, using the Support Vector Machine (SVM) classification algorithm, Tseng et al. were able to classify a sperm cell as normal or abnormal based on sperm head morphology features obtained by extracting contour lines.⁷¹ More importantly, abnormal sperm can be classified into different groups based on the type of the morphological abnormalities (e.g. tapered head, round head, pyriform, and others). Such information is valuable for the treatment of infertility cases where the number of abnormal sperms is high compared to that of normal sperm. Chang et al.⁷² compared the classification performance of four supervised learning algorithms (1-Nearest Neighbor, SVM, decision tree, and naïve Bayes) using a sperm head morphology data set composed of 1872 labelled images (see Figure 2 for examples of SVM

and decision tree). These images had been labeled by three expert clinicians as normal or as one of four abnormal classes: tapered, pyriform, small, and amorphous. The whole dataset was divided into a training set (60%), a validation set (20%) and a testing set (20%). For training each of the four classification algorithms, each sperm head was defined using a shape descriptor (either *Hu* moment, Zernike moment or Fourier descriptor based on mathematical operation) that could numerically express different head features such as area, perimeter and circularity. This shape descriptor was then correlated with the labelled (i.e. known) classification for each image to complete the training phase. The prediction performance of the algorithm was then tested using 20% of the images. Among the three descriptors and four classification algorithms, the combination of Fourier descriptor with SVM performed the best with 49% prediction accuracy. The authors also reported a two-stage classification algorithm in which the first stage discriminates amorphous sperm from the rest and then the second stage classifies the sperm into the four abnormal classes.⁷³ The two-stage classification algorithm achieved an average true positive rate of 58% using a dataset in which two out of three experts agreed on a classification. However, even in the more stringent data subset in which all three experts agree on a classification, the true positive rate increased only to 73%. The authors attributed the mild improvement to the variability in assessing a sperm between the experts – that is, the quality of the training data.⁷³

Microscope images of sperm were also used directly in ML for morphology classification. Shaker et al.⁷⁴ used a patch-based dictionary learning method wherein sperm images labeled with a class were divided into patches to identify important features in the sperm (Figure 2c). The authors demonstrated an improved average true positive rate of 62% compared to Chang et al.'s work⁷² using the same data set. They also achieved an average true positive rate of 92.3% using a new data set in which all three experts agreed on the sperm class for all cases.⁷⁴ Instead of

characterizing each sperm with extracted features and shape descriptors from the image, morphology of a live and motile sperm cell can be obtained from video data to classify sperm. Van Raemdonck et al.⁷⁵ reported an algorithm that can classify $\sim 10^2$ motile sperms into normal, tapered, round, small and pyriform categories according to the measured properties such as area and length from sperm head images. However, the low spatiotemporal resolution of sperm head images from video data likely influences the accuracy of the algorithm. Moreover, background noise in the images due to other cells and debris in the raw semen sample also had an impact.

Deep learning has gained considerable popularity for sperm classification in recent years due to its unique capability to extract information directly from images. Deep learning provides an extra level of prediction accuracy for sperm morphology classification compare to other ML approaches by eliminating potential errors that can arise during feature extraction. Recently, a CNN was applied to classify sperm. A CNN is a class of deep neural network (DNN) for analysing images that is very effective in image recognition or classification. An algorithm called Visual Geometry Group (VGG) 16, which had been pre-trained on human-annotated stock images in the ImageNet database, was retrained to classify sperm.^{49,76} Using VGG 16 and the same data set used by Shaker et al.⁷⁴, Riordon et al.⁴⁹ achieved a true positive rate of 94%, outperforming Shaker et al.⁷⁴ and performed as well as the patch-based dictionary learning method by Chang et al.⁷² They also eliminated the excessive computation required for training by using a pre-trained deep learning network. Javadi and Mirroshandel also showed that deep learning is effective in assessing low-resolution sperm head images,⁷⁶ demonstrating its potential application in real clinical settings where the real-time use of high-resolution microscopy is limited.

These ML methods complement, and in many cases improve upon, current methods by standardizing and expediting the evaluation of sperm. When combined with emerging cellphone-

based imaging technologies for sperm analysis,⁷⁷ these technologies provide promising opportunities for at-home male fertility screening. Due to the large amount of data required for training a ML algorithm, better collaboration between fertility clinics and research laboratories is needed. Additionally, improvement in areas such as digital imaging and image processing are preferable, particularly to enable high-resolution and high-speed imaging of motile sperm that can improve the prediction capabilities of current ML-based morphology analysis approaches.

Machine learning strategies using DNA integrity and sperm motility

To develop an ML algorithm that can be implemented clinically for sperm selection, sperm must be assessed not only on morphology but also on motility and DNA integrity. The natural selection process *in vivo* is comprehensive, providing a spectrum of rheological, biochemical and geometrical barriers to the sperm sample as a whole.^{78,79} However, current practices for sperm selection in ART involve pre-screening based loosely on the motility, followed by a final inspection of a few sperm based on morphology.²⁴ For example, if a subpopulation of sperm is selected using DG or SU, the sample is still widely distributed in quality, and not all of sperm in the selected population demonstrate the motility, normal morphology and DNA integrity suitable for fertilization. The embryologist manually selects sperm from that sub-sample, without any ability to assess, for instance, low DNA integrity that is associated with infertility.⁸⁰ Although current ML-based approaches for morphology assessments opens possibilities for standardization, developing ML algorithms for sperm motility and DNA integrity analysis at the single sperm level is essential to achieve a comprehensive selection process. Nevertheless, studies on single sperm DNA integrity and possible correlations with motility and morphology have been limited.

Assessment of DNA quality and its use for training ML algorithm

Current methods of sperm DNA analysis are both invasive and limited to the quantification of DNA fragmentation at the sample level. Although there are techniques that could provide quantitative measurement of individual sperm's DNA, the results are usually interpreted as an average value of a few hundred or a few thousand cells. The information obtained from these sample-based measurements does not directly inform on the quality of a single sperm of interest, and thus is of limited use in final sperm selection.⁸¹ Here, ML provides promising avenues to develop quantitative methods for non-invasive sperm DNA analysis at the single sperm level and to develop ML algorithms for sperm selection with single-cell resolution.

Currently, four types of DNA fragmentation assays are mostly used: sperm chromatin structure assay (SCSA),⁸¹⁻⁸³ terminal deoxynucleotidyl transferase UTP nick-end labeling (TUNEL),⁸⁴ sperm chromatin dispersion (SCD),⁸⁵⁻⁸⁷ and single cell gel electrophoresis (SCGE) or the comet assay (Figure 3a-e).⁸⁸⁻⁹¹ The underlying mechanism in each is the fluorescent staining of the DNA and subsequent quantification of the fragmentation. The invasive nature of the staining step precludes the direct application of DNA assessment on candidate sperm for IVF or ICSI. That is, if a cell is stained, that cell cannot be used clinically, but it *can* be used to train an algorithm to establish a correlation between the morphology and the DNA integrity of a single sperm. For SCSA and TUNEL, the fragmented DNA and intact DNA appear in two different colours. In the case of SCD and comet assay, the transport of DNA molecules through a gel is visualized and quantified using a single staining or fluorescent dye. While all the assays can evaluate sperm DNA fragmentation at the sample level, there is still a need to develop a reliable non-invasive method to study the DNA fragmentation at the single sperm level. Although the comet assay provides a

quantitative measurement at the single sperm level, the variability due to unstandardized protocols is a challenge in the application of this method.⁹²

The first attempt to establish a correlation between DNA integrity and sperm morphology at the single sperm level was conducted by Garolla et al.⁹³ In this study, sperm selected manually based on morphology using high-magnification microscopy were analyzed for DNA integrity. The results showed significantly improved mitochondrial function and chromatin status for morphologically selected sperm as compared to the unselected population. Utsuno et al. also explored the correlation between sperm head shape and DNA fragmentation for single sperm, demonstrating a higher rate of DNA fragmentation for sperm with abnormal head shapes.⁹⁴ After localizing an individual sperm, high-magnification images were used to correlate ellipticity, symmetry, and angularity of the sperm head shape with the DNA integrity of the same sperm, as assessed by the TUNEL assay. However, the sample size was relatively small, with 125 sperm from 10 donors. Recently, Wang et al. demonstrated a method for assessing the DNA fragmentation index (DFI) of individual sperm cells using the acridine orange (AO) staining method (Figure 3f-g).⁹⁵ After immobilization of sperm on a hyaluronic acid (HA) modified glass slide, the morphology and DNA integrity of the same ~1000 sperm were analyzed. Correlations between quantitative morphological parameters and the DFI of single sperm were established and used to predict single sperm DFI from morphological parameters. Using a linear regression model, the weight of morphological parameters contributing to the sperm DFI were estimated, and logistic regression was used to provide a more accurate prediction of single sperm DFI. The work demonstrated for the first time a systematic analysis of the correlation and prediction of sperm DNA integrity based on morphological parameters at the single-cell level.

Following this work, McCallum et al. developed a deep learning algorithm that can predict the DNA integrity of sperm based on the morphology.⁹⁶ The algorithm is fed sperm images and the DNA fragmentation index (DFI) is predicted (Figure 3h). This work was the first demonstration of a ML algorithm that can determine, without human bias, the quality of individual sperm cells, solely based on sperm images. This work demonstrated the possibility of standardizing the single sperm morphology based on DNA fragmentation, a quantitative measure free from clinician subjectivity.

The approach taken by McCallum et al. can serve as a basis for including other sperm characteristics such as protein expression for training a smarter algorithm. For instance, expression of specific proteins such as HSPA2⁹⁷ or IZUMO1⁹⁸ associated with sperm-egg binding and fusion, respectively, can also be used as quantitative measures to train an algorithm, or to extend the training of the algorithm originally trained with DNA fragmentation data.

Assessing the motility of individual sperm

For natural fertilization, sperm must navigate through the female fallopian tube towards the oocyte, making motility essential. Motility also plays a key role in shaping the decisions of an embryologist when selecting a sperm for ICSI. Currently, motility is analyzed at the sample-level using computer-assisted sperm analysis (CASA), providing useful information of the sample such as VAP (average path velocity), VSL (straight line velocity), VCL (curvilinear velocity), ALH (amplitude of lateral head displacement) and BCF (beat cross frequency).⁹⁹ Recently, these factors were used to train a SVM algorithm to classify sperm motility.^{100,101} Although the CASA system can provide motility information at the sample-level, it is difficult to determine the motility of a specific sperm and match that information to other qualities such as morphology or DNA integrity.

Deep learning provides new opportunities to predict sperm motility and further link it to other sperm quality metrics in the service of sperm selection. Hicks et al. trained a CNN using videos of semen samples combined with participant data (i.e. age, body mass index (BMI), and days of sexual abstinence) to predict the motility data and classify the sample as progressive, non-progressive or immotile.¹⁰² However, no noticeable difference between the prediction performance of the algorithm was observed between the case with and without participant data. Therefore, motility data on single sperm must first be obtained in order to fully leverage the prediction potential of a ML algorithm.

Over the years, many methods of studying the motility of single sperm via tail beating were suggested. The sperm head fixation method (SHFM) developed by Hiramoto and Baba¹⁰³ has been frequently used to hold individual sperm to study its tail beating (Figure 4a). Keeping the head at the same position enabled detailed analysis of tail beating, providing important information such as beating frequency and amplitude, which eventually led to mathematical models that explain the tail beating characteristics of a sperm.¹⁰⁴ Advances in high speed cameras have also allowed high resolution, *in-situ* recording of swimming sperm without the need for fixing the head on a substrate. This resulted in the development of computer software such as SpermQ¹⁰⁵ or FAST¹⁰⁶ that enable tail beating analysis of a swimming sperm. Holographic imaging has also been used for tracking individual sperm and analyzing motility and tail beat (Figure 4b-f).^{107–109} These techniques can provide detailed information on sperm tail beating (e.g. waveform, amplitude, beating frequency) and demonstrate the possibility of combining such techniques with ML to develop powerful algorithms. Dubey et al. recently reported a digital holographic microscopy method combined with machine learning for quantitatively analyzing the influence of oxidative stress on the morphology and motility of sperm.¹¹⁰ Methods and techniques such as this can provide grounds for establishing

correlation between different sperm parameters such as morphology and motility at the single cell level. Also, in recent years, microfluidic platforms have been used to develop devices to isolate individual sperm to analyze the tail beating frequency and amplitude (Figure 4g-j).¹¹¹⁻¹¹³ In particular, You et al.¹¹³ developed a high-throughput platform that can assess the morphology, tail beating and DNA integrity of individual sperm using an array of 2400 traps (Figure 4j). This platform can establish the link between the morphology, tail beating and DNA quality of individual sperm, providing opportunities for training a ML algorithm.

Once the motility information, such as tail beat, is combined with metrics of morphology and DNA integrity, the ensemble data can be used to train an algorithm. Such an ensemble of sperm data would provide an opportunity for ML models to comprehensively inform clinical practices of individual sperm selection by predicting highly quantitative measurements and outcomes (DNA quality and fertilisation rate) from clinically accessible image inputs (representing shape, size, intensity values).

Training an algorithm with motility data will require image sequences or videos as input – a task well suited to a recurrent neural networks (RNNs) (Figure 4k). In RNNs, videos are processed as sequential image data, selectively passing information across sequence steps powered by long short-term memory (LSTM).^{46,114} RNNs are very deep networks for which all layers share the same weights, which rely heavily on backpropagation and LSTM units in order to mitigate diminishing gradients and to learn long-term dependencies. In particular, the LSTM introduces a composite memory cell with specifically connected simpler nodes to overcome vanishing or exploding gradients and learn long-term features. To date, RNNs have been used primarily for text processing^{115,116} or image and video analysis.¹¹⁷⁻¹¹⁹ An RNN can output either a single value (e.g., the sentiment of a sentence of text) or multiple values (e.g., the translation of text into another

language or the classification of each frame in a video). Recent applications of this type in biology generally include the discrimination and prediction of cell motility patterns¹²⁰ and the detection of DNA base modifications from gene sequencing data.¹²¹ These approaches highlight the power of RNNs to glean high level information and improve clinical outcome prediction. For fertility applications, RNNs exploit the sequential nature of the data and can be trained to automatically extract the tail-beat pattern and motility of individual cells by analyzing differences in individual image frames. Such an algorithm could be trained in current clinical practice, on clinical ICSI selection videos for which there is excellent outcome data including blastocyst rates, embryo development rates, and even pregnancy outcomes - thereby directly correlating sperm video data with quality objective metrics. Additionally, similar to CNNs, RNNs can be trained to predict cell DNA integrity from images. RNNs hold even more potential in this regard because the algorithm can exploit both the embedded motility information present in the image sequence as well as the data-rich image frames. The primary limitation thus far to RNN-based sperm video analysis is data acquisition since individual cells must be tracked and recorded and subsequently stained to assess the DNA integrity. One approach to this challenge is to organize the cells in a way conducive to imaging and analysis, such as the parallel microfluidic sperm traps developed by You et al.¹¹³ Ultimately, once the algorithm is fully trained, it can be combined with an online tracking method such as the one shown by Zhou et al.¹²² for implementation in the clinic.

Additional considerations for the application of ML for sperm selection

Many advances have been made in the realm of machine learning for fertility applications and for medicine in general. However, several factors still need to be taken into considerations including the influence of data bias or the type of model to train the data with. More importantly,

ethical questions need to be asked and addressed before the implementation of machine learning for sperm selection.

Effect of data bias

The potential for data bias is an important consideration in any application of machine learning. In the context of sperm selection, several aspects of data such as quality, subjectivity, representativeness, and size could lead to bias.

Data quality

The quality of data is critical for machine learning, especially when dealing with image data.^{123,124} Here, the image data refer to microscope images or video frames of sperm acquired using a microscope during semen analysis or during sperm selection in the ICSI procedure. Generally, higher resolution images provide more information to the algorithm and lead to better performance. Although classification of low-resolution and noisy images via a deep learning model was demonstrated⁷⁶, most algorithms are trained with high resolution images ($\geq 60\times$ objective lens). Earlier research has demonstrated greater success when classifying sperm imaged with a $100\times$ objective lens⁷⁴ relative to a $63\times$ lens⁷². However, this level of resolution may not be suitable for clinical application which requires lower magnification for adequate field of view and motility assessment. Specifically, to implement ML-based sperm selection technology within the ICSI workflow, the algorithm needs to be trained with images and videos of sperm at the same resolution used for the procedure (though training with higher quality images may also have a role – see *Future Directions*). Although intracytoplasmic morphologically selected sperm injection

(IMSI) has been developed to use high-resolution imaging to select sperm, no beneficial effect has been found compared to standard ICSI.¹²⁵

In addition to image resolution, imaging conditions such as exposure, brightness, and uniformity of lighting are factors that can affect prediction results.¹²⁶ The performance of ML algorithms is always bound by the quality of the training data. Advances in microscopy and imaging techniques have provided potential solutions to such issues. For instance, Dai et al. reported an automated non-invasive imaging system for studying the morphology and motility of single sperm.¹²⁷ This system is capable of relocating the sperm after increasing the magnification, enabling the acquisition of both motility and morphology without sacrificing the image quality.

Subjectivity in training data

When training a ML algorithm for sperm selection, bias will also arise if the training data is labeled subjectively by embryologists. As in morphology screening, interpretation of normal sperm is not consistent for every embryologist performing the selection.⁷² Many works^{54–56} report inter and/or intra-laboratory variability as well as the influence of differing levels of experience among clinicians performing sperm analysis as having an effect on the labeling of data. Moreover, with the current lack of a quantitative standard that can differentiate the very best among all sperm, the issue of subjective interpretation will be an even more difficult challenge to overcome. As the results of an algorithm will only be as good as the quality of the training data set, variability in the labeling of data will likely lead to biased predictions by the algorithm⁴²

Training with objective single sperm characteristics (i.e. DNA integrity, genomics, and proteomics) could address the issue of subjectivity in labeling sperm by clinicians. Establishing and utilizing correlations between many sperm metrics (morphology, motility, DNA quality, and

others) can also mitigate the issue of subjectivity in individual metrics.¹²⁸ Augmenting training with protein or gene expressions can further reduce bias and improve standardization.¹²⁹

Although training with a wide spectrum of accepted sperm quality metrics is the recommended approach, it is noteworthy that all such metrics are simply accessible proxies for the ultimate outcomes of interest: fertilization rate, live birth rate, and the health of offspring.

Representativeness and size of the data set

Even training data sets based on objective metrics of sperm quality can be inherently biased in terms of representation. For instance, if the samples are obtained only from healthy males, or males from a particular age group or ethnicity, then there is a risk of bias within the data set. Gijssberts et al.¹³⁰ reported that a model trained with data on cardiovascular risk factors based on a primarily white population led to both overestimation and underestimation of risk of cardiovascular events in non-White population. Similarly, researchers found that commercial facial recognition software with racial imbalances in training data misidentifies under-represented groups such as darker-skinned females with error rates up to 34%.¹³¹

Potential differences of sperm quality based on demographic factors also need to be considered. Various research groups have found significant differences across different demographics.^{132–134} Men from different socioeconomic backgrounds may have varying levels of semen concentration, volume, or sperm motility, which has clinical standardization ramifications. Whether demographic and socioeconomic factors influence the sperm quality at single cell level remains an open question. Nonetheless, the algorithm should be trained with data from a wide range of samples from men spanning demographic and socioeconomic backgrounds to avoid potential bias.

The need to achieve a sufficiently representative, objective, and high-quality data set influences the sample size. Although data from 100 male patient samples is more likely to be biased than data from 10,000 samples, determining the minimum sample size is a persistent challenge. In practice, there are trade-offs to be made and researchers need to choose between less data of higher quality versus more data of lower quality. Likewise, the number of sperms analyzed per sample versus the number of samples to analyze is another practical decision that must be made in the absence of clear guidelines.

Machine learning model considerations

Overfitting, data class imbalance, and generalization are important considerations in the training of algorithms. Overfitting is commonly mitigated on the data side through image augmentation or on the network side via cross-validation and regularization methods such as dropout and batch normalization. Imbalance in data classes – when the quantity of training samples is vastly different across different categories – can be mitigated by weighting low-quantity data higher or by copying and augmenting low-quantity samples. Lastly, generalization concerns (i.e., how well a trained algorithm can be applied to a new dataset) can be addressed by cross-validation. Ultimately, an algorithm should only be used in practice once proven effective and independently verified.

Ethical concerns

Machine learning in sperm selection is an emerging area of research with many associated ethical concerns yet to be fully understood and addressed. A framework with which to assess these ethical issues is emerging from the inputs of experts in legal, medical and technology sectors.^{135–139}

The application of ML in sperm selection shares many ethical considerations and implications with other areas of medicine such as disease diagnostics^{140–142} and medical imaging.^{143,144} In many

cases, detailed ethical framework lags the technology. For instance, efforts are emerging to commercialize AI-driven embryo analysis and selection as exemplified by Life Whisperer¹⁴⁵ and Future Fertility¹⁴⁶, while the ethical and legal framework around this technology is yet to be fully developed.¹⁴⁷ Below are ethical concerns relevant to ML-driven sperm selection technology.

Over-reliance on ML and liability

When a medical decision is made, the clinician is responsible for its consequences.¹³⁷ Although the diagnosis and patient treatment decisions made by clinicians are based on medical evidence, many of them result in undesired diagnostic error. A survey performed on practicing internal medicine doctors from nine different clinics in the U.S. revealed that more than 40% of the doctors make at least one uncertain diagnosis daily.¹⁴⁸ With recent developments of ML technologies, the medical community is hopeful that such technologies would assist clinicians in mitigating diagnostic errors.

How heavily should a clinician's medical decision rely on algorithm input? More specifically, if the clinician and the algorithm have contradicting opinions – for instance on the appropriateness of selecting a given sperm – who (or which) should make the final decision? If it is the algorithm that makes the decision, who is responsible in case of unwanted consequences? These questions, as in other areas of medicine, need to be addressed before implementation. Currently, the law only holds clinicians liable when they do not follow the standard of care when making a medical decision.¹³⁷ However, if medical decisions rely heavily on ML algorithms, the procedure and the extent of clinician involvement may be questioned.¹³⁵

Given these complexities, it is likely that in the near- and medium-term, ML will be introduced as an assistive technology only, helping the clinician make the final decision, but not operating

without clinician input.¹⁴⁹ Other medical fields such as neurosurgery^{149,150} or radiology^{151,152} have started to look at this option more closely - either implementing it in clinical practice or evaluating its potential through clinical research. For instance, according to a global survey by Staartjes et al., it was found that ~29% of surgeons employ ML in their clinical practice mainly for prediction of outcome and complications.¹⁵⁰

Data privacy

Who owns the data collected by ML algorithms used during training? For any ML application, data privacy is an important topic that must be considered. For training a ML algorithm that is generally applicable, an abundance of data on single sperm from many individuals will be required. While this may be less of an issue for an algorithm trained only on sperm characteristics, there would be added risks if more subject-specific parameters were involved including age, body mass index, or weight are included.

As algorithms improve, sharing of personal information including age, mass body index, weight, ethnicity, or semen quality may be needed to accommodate the large amount of data required to train a smarter algorithm.¹⁵³ For instance, VerMilyea et al. developed a deep learning algorithm for the prediction of embryo variability based on images of 8886 embryos obtained from 11 different clinics across three different countries.¹⁴⁵ Similarly, Tran et al. also showed an algorithm for predicting fetal heart pregnancy using images of 10,638 embryos from four different countries.¹⁵⁴ In such cases, de-identification of image/video data is critical to ensure the privacy of the patients. In addition, even if the data are de-identified, they must not be available to the public. Finally, patients should be explicitly told about data sharing and obtaining consent prior to the use

of data for research. The security of such systems must be ensured to protect the personal information of patients.¹³⁸

Variability and diversity

While current clinical practices are standardized to some extent, most practitioners recognize that there is a significant degree of variability in sperm selection processes from clinic to clinic, clinician to clinician, and even the same clinician over time. While increased standardization could improve outcomes overall, there may be a risk of over-standardization. It is conceivable, for instance, that ML-guidance would favour a particular subpopulation, more so than the natural process or the current clinical practice. The reality of current clinical practice is that sperm selection practices are highly variable. While reducing this variability with consistent application of ML may be beneficial in many regards, it is also possible that the current clinic-to-clinic and clinician-to-clinician variability are – on the whole – good for the population. This remains an open question.

Future Directions

In order to realize a clinical ML algorithm for sperm selection, there are still several steps ahead. First, being able to acquire data on live, freely suspended sperm is a priority in order to implement reliable ML approaches for sperm selection. As discussed above, most of the reported ML approaches use high-resolution, fixed and/or stained images of sperm to train an algorithm. However, it is difficult to directly use these algorithms in clinical settings since sperm are mobile and imaging resolution is not as high. Recent developments in advanced imaging methods may offer potential solution to this issue. Approaches such as a high-speed off-axis holographic

system¹⁵⁵ or the automated magnification switching method¹²⁷ can be useful for obtaining high-resolution images of moving sperm. With proper implementation of these systems in clinical workflow, ML approaches could be made applicable to a clinical setting.

Alternatively, low-resolution images (and videos) captured in a clinical setting could potentially be improved using Single Image Super Resolution (SISR). In SISR, a high-resolution counterpart of a low-resolution image is produced by one of several methods, including interpolation^{156,157} and example-based ML.¹⁵⁸ With the development of deep learning algorithms, the SISR has been used in various biomedical applications such as improving MRI images for tumor detection¹⁵⁹ or image resolution for endomicroscopy.¹⁶⁰

Along with issues surrounding imaging, a standardized quantitative assessment of sperm morphology must be established. The WHO criteria are subjectively interpreted by clinicians during sperm selection for ICSI. However, classifying sperm simply as normal and abnormal is insufficient. For any ML approach to be implemented for sperm selection, the correlation between single sperm morphology – preferably obtained from live, freely swimming sperm – and its fertilization potential must be clearly established. Assessment of a single sperm's fertilization potential may be accomplished by quantifying DNA integrity at the single sperm level, in addition to using other quality markers, such as testing for the presence of chromosomal aneuploidy or measuring the expression of certain proteins. For instance, a ML algorithm trained with morphology data labeled with the presence/absence of chromosomal aneuploidy in a single sperm could help clinicians avoid selecting such sperm during ICSI (if such a prediction is possible). Therefore, sperm metrics at the single cell level not only offer many opportunities for the scientific community, but it is also crucial to realizing a “smart” ML algorithm that can evaluate a sperm's fertility potential.

The standardization of single sperm assessment by establishing a correlation between morphology and other parameters can benefit from the imaging technologies discussed above. Also, these aspects require a broad collaboration of the clinical and engineering communities and the sharing of ML-accessible sperm data on which to train algorithms – an open-source approach to the fertility challenge – powered by fundamental research on single sperm parameters and the advancement of tools to facilitate such efforts.

Selection criteria for cited articles

The original articles were selected based on the relevance to the topics discussed in the manuscript, namely sperm morphology assessment, DNA integrity, sperm motility, and machine learning. Articles of interest were those describing, evaluating, and discussing sperm morphology and DNA integrity assessment techniques as well as those that propose the use of machine learning for sperm classification. As most current sperm morphology and DNA assessment approaches are based on techniques developed several decades ago, reports with publication from 1978 to 2020 have been referenced in the review. In the ethics section, we have referenced specialized perspectives and review papers discussing the ethical issues of artificial intelligence and machine learning written by experts in legal, medical, and technological sectors.

Conclusion

The application of machine learning is rapidly expanding within biomedical applications. The capability of this approach to process and analyze large amounts of data – and particularly image data – will be of particular utility in sperm selection. Adequately trained ML algorithms could enable embryologists to improve and standardize sperm selection, which could increase the

fertilization success rate and outcomes more generally. Developing such an adequately trained ML system will require great attention to data quality, objectivity of metrics, representativeness, and sample size, as well as addressing ethical questions arising from the increased mechanization of reproduction.

Display Items:

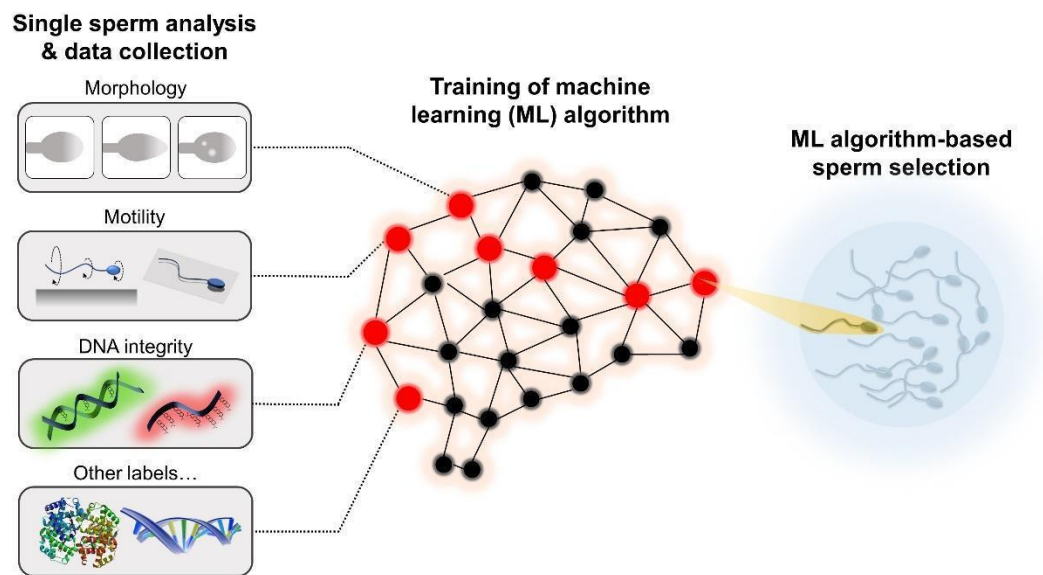


Figure 1 | The process flow for a ML algorithm for sperm selection. Morphology, motility, DNA integrity, other sperm quality metrics can be used to train a ML algorithm for sperm selection with single-cell resolution. The potential for ML in sperm selection motivates a rapid acceleration in the study of single-sperm quality.

Table 1| Timeline of WHO morphology assessment criteria. Over time, the morphology criteria have become stricter with cut-off values from 80.5% to 4% (adopted with permission from ref. 53)

WHO edition	Year	Criteria	Methods of assessment	Cut-off for normal forms (%)	Calculation of cut-off values
1 st	1980	Liberal approach	No clear description of normal forms, 'classification based on MacLeod's description'	80.5	Mean of fertility population (range = 49-98%)
2 nd	1987	Liberal approach	No important difference compared with the 1st edition	≥50	No precise data
3 rd	1992	Strict criteria	Clear description of normal spermatozoa according to strict criteria with well-defined sperm head lengths and widths and qualitative descriptions	≥30	Arbitrary value
4 th	1999	Strict criteria	List of various abnormalities without accurate description	14	No precise value given, 'multicenter studies refer to >14% for IVF'
5 th	2010	Strict criteria	Precise definition of normal spermatozoa and of different abnormalities	4	Lower reference limit (lower fifth centile value), data from fertile men

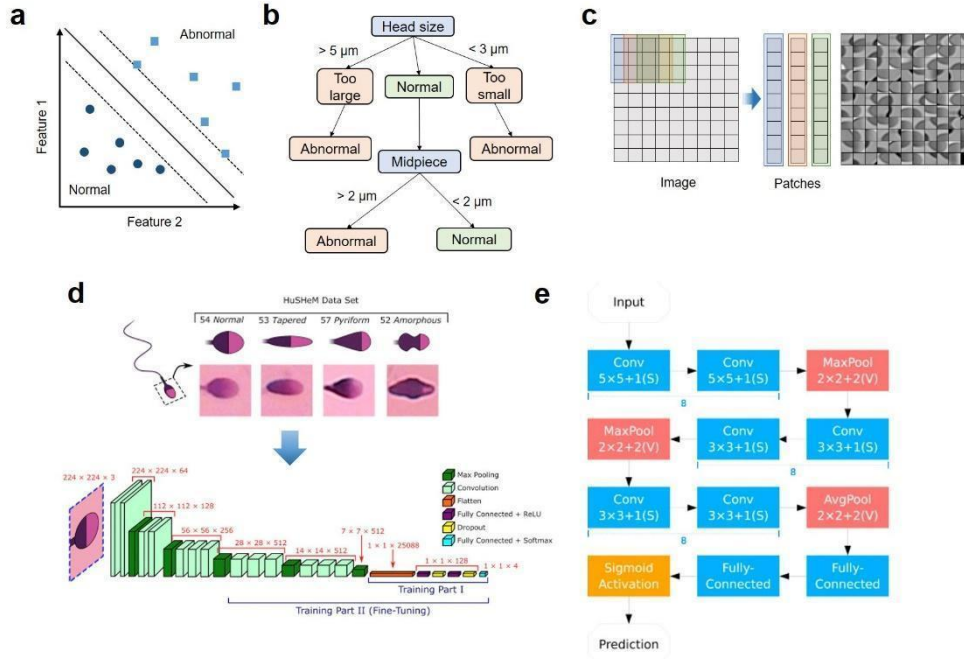


Figure 2 | Machine learning for sperm selection. Machine learning algorithms such as **(a)** Support Vector Machine (SVM) classifier, **(b)** Decision tree, and **(c)** Patch-based dictionary method have been applied to classify sperm.⁷⁴ A SVM is mostly used for a two-group classification where the groups are divided based on a decision boundary vector. Reproduced with permission from ref. 74. On the other hand, a decision tree is composed of a series of nodes where a binary classification (Yes or No) is made. **(d, e)** deep learning approaches have also been used to classify sperm based on label given by clinical experts. Adapted with permission from ref. 74 and 76, respectively.

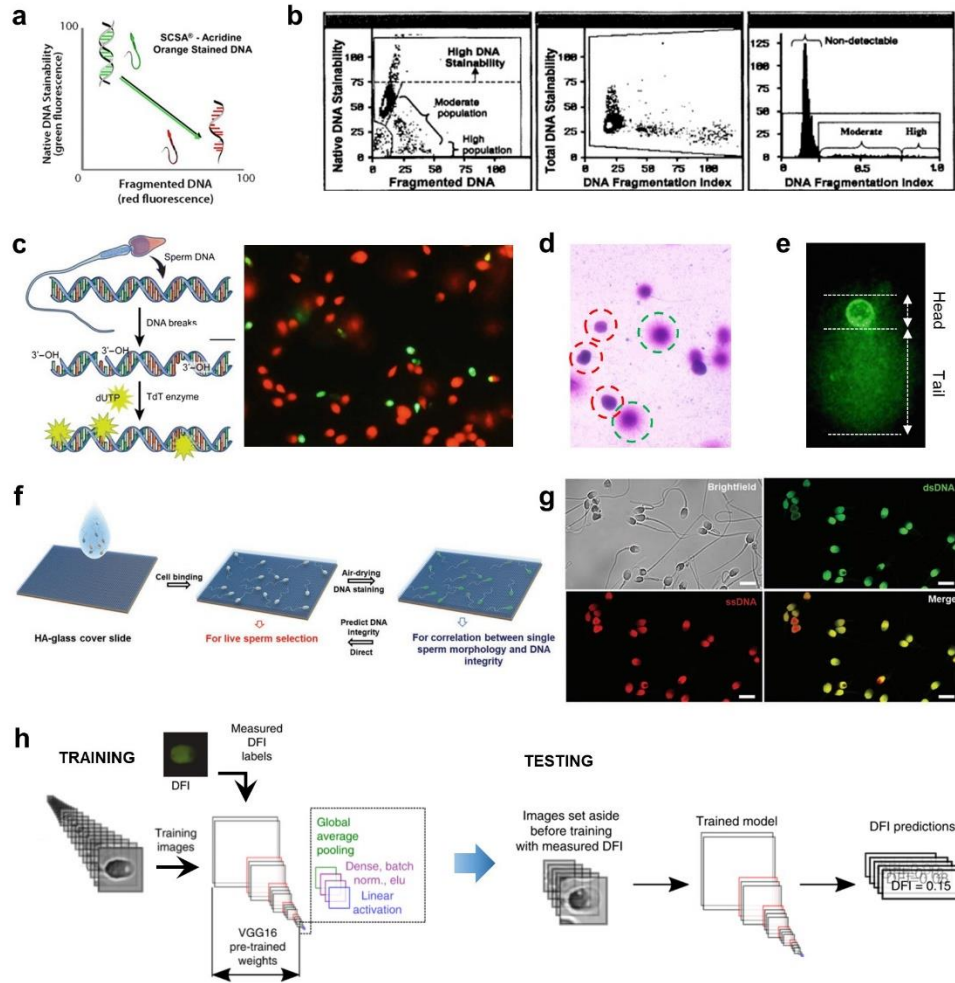


Figure 3 | Different methods of analyzing the DNA integrity of a sperm sample. (a) The sperm chromatin structure assay (SCSA) is the most widely used method for analyzing the average DNA integrity of a semen sample.⁸¹ Reproduced with permission from ref. 81 (b) In SCSA, the DNA fragmentation index is calculated by thresholding the data obtained from flow cytometry.⁸¹ Reproduced with permission from ref. 81 (c) The terminal deoxynucleotidyl transferase UTP nick-end labeling (TUNEL) assay tags the sites of DNA breaks using fluorescent-labeled dUTP, rendering the sperm with DNA fragmentation green in colour while sperm with no (or low) DNA fragmentation appears red. Reproduced with permission from refs. 161 and 162. (d) Sperm chromatin dispersion (SCD) tests for DNA-fragmented sperm allow DNA fragments to diffuse through agarose gel thus preventing them from being stained. Therefore, sperm with fragmented DNA show small (or no) halo (circled in red colour). On the other hand, sperm with intact DNA gets stained with the dye displaying a large/medium halo around the cell head (circled in green colour) Reproduced with permission from ref. 163 (e) Single cell gel electrophoresis (SCGE) or comet assay use a similar principle, but fragmented DNA is forced to travel through agarose gel by electrophoresis while the intact DNA remain in place. Reproduced with permission from ref. 88 (f)-(g) Recently, the single sperm DNA fragmentation index (DFI) was measured and correlated to individual sperm morphology. The method was enabled by immobilizing the sperm cells on a hyaluronic acid modified substrate. Reproduced with permission from ref. 95 (h) Using the single sperm head images with corresponding DFI values, a deep learning algorithm was trained to predict the DFI of individual sperm based on sperm morphology. Reproduced with permission from ref. 96.

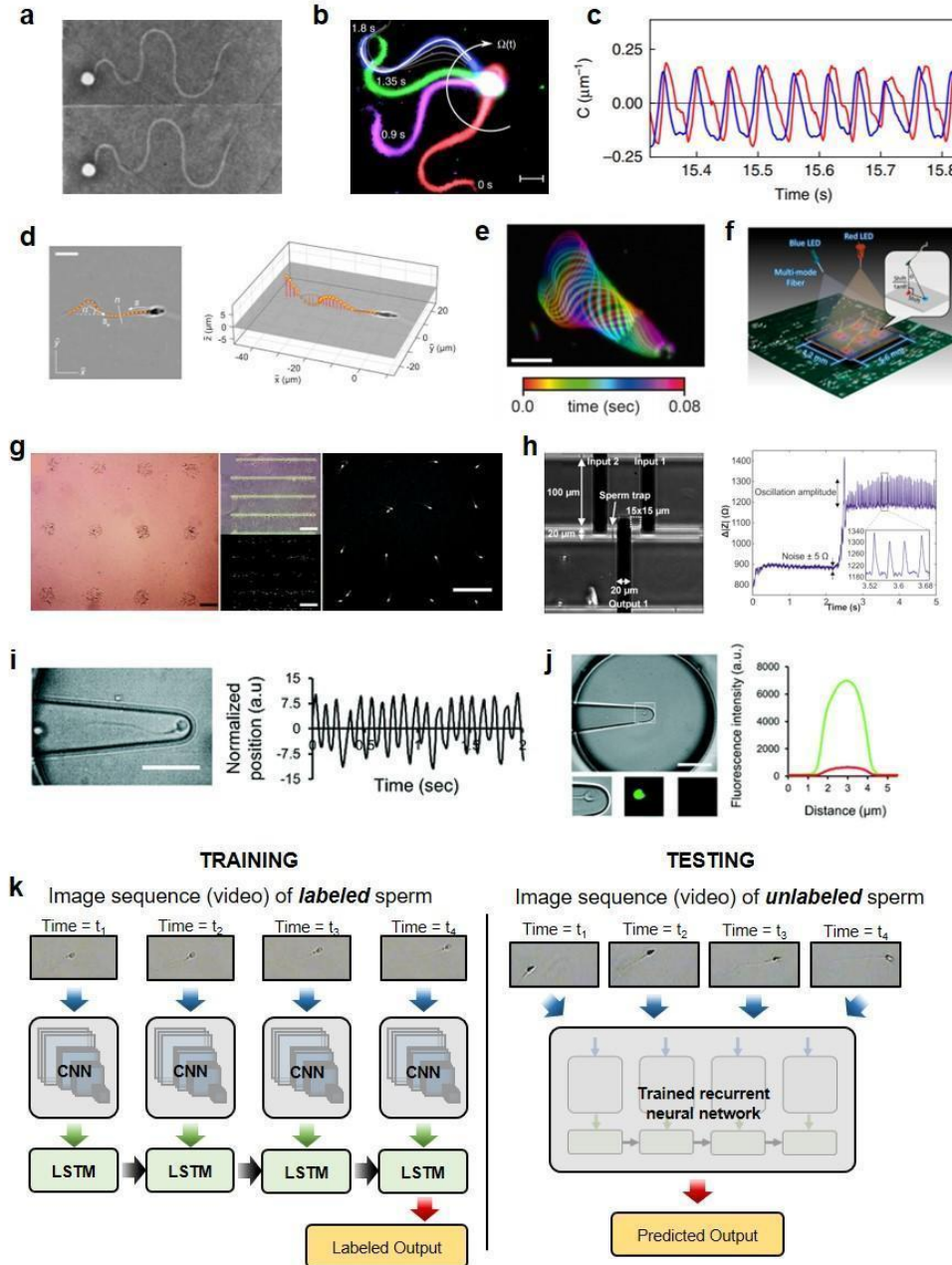


Figure 4 | Approaches for sperm motility analysis. (a) Sperm head fixation method (SHFM) first introduced by Hiramoto and Baba. Reproduced with permission from ref. 103. (b) With advances in imaging techniques, sperm tail beating can now be tracked with high spatiotemporal resolution. Reproduced with permission from ref. 104 (c) With the head fixed to the substrate, sperm exhibit a sinusoidal tail beating waveform. Reproduced with permission from ref. 104 (d, e) The union of high-speed recording technology and computing power has enabled the analysis of tail beat in free swimming sperm. Reproduced with permission from refs. 105 and 165. (f) Holographic imaging can also track individual sperm during navigation. Reproduced with permission from ref. 108 Platforms that enable the analysis of many individual sperm tail beating have also been introduced: (g) Fibronectin dot patterns for single sperm array (reproduced with permission from ref. 111, (h) a microfluidic platform for trapping individual sperm for impedance-based sperm tail beat analysis, (reproduced with permission from ref. 112) and (i) a microfluidic trap for tail beat analysis and high-resolution sperm imaging (reproduced with permission from ref. 113) (j) In particular, the microfluidic trap also enables the DNA analysis of individual sperm after analyzing tail beating (reproduced with permission from ref. 113) (k) Scheme showing how sperm motility data in either video or image sequence format can be used for training of recurrent neural network (RNN) for live sperm selection.

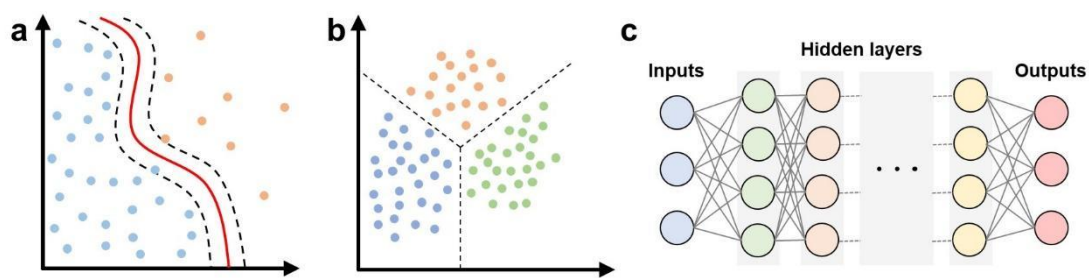


Figure B1 | Schematics showing examples of machine learning strategies (a) Classification (b) Clustering (c) artificial neural network

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Author contributions

J.B.Y., C.M., Y.W., J.R., and R.N. researched data for and wrote the article. D. S. supervised the work, wrote and revised the article. All authors made substantial contributions to discussion of content and reviewed and edited the manuscript before submission.

Competing interests

The authors declare no competing financial interests.